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MATHEMATICS 2270
Midterm 3, Fall semester 2012

Name: Key

1. (20 pts) Diagonalize the matrix $A = \begin{pmatrix} i & 1 \\ 1 & i \end{pmatrix}$ and compute A^n .

$$\begin{vmatrix} i-\lambda & 1 \\ 1 & i-\lambda \end{vmatrix} = \lambda^2 - 2i\lambda - 2$$

$$\lambda = \frac{2i \pm \sqrt{-4+8}}{2} = \frac{2i \pm 2}{2} = i+1, i-1$$

$$\lambda_1 = i-1 \quad N \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \left\langle \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\rangle$$

$$M = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \quad M^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$\lambda_2 = i+1 \quad N \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \left\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle$$

$$M^{-1}AM = \begin{pmatrix} i-1 & 0 \\ 0 & i+1 \end{pmatrix} = \Lambda$$

$$A^n = M \Lambda^n M^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1^n & \lambda_1^n \\ -\lambda_2^n & \lambda_2^n \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \lambda_1^n - \lambda_2^n & \lambda_1^n + \lambda_2^n \\ -\lambda_1^n - \lambda_2^n & -\lambda_1^n + \lambda_2^n \end{pmatrix}$$

ok for full credit

2. (15 pts) Find a linear map that sends the column vectors of $\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$ to the column vectors of $\begin{pmatrix} i+1 & i-1 \\ 1 & 1 \end{pmatrix}$.

$$\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} i+1 & i-1 \\ 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}^{-1} \quad \begin{pmatrix} i+1 & i-1 \\ 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} i+1 & i-1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}^{-1} = \begin{pmatrix} i+1 & i-1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -i & -1 \\ -i & 1 \end{pmatrix} \cdot \frac{-1}{2i} = \begin{pmatrix} -i^2 - i - i^2 + i & -i - 1 + i - 1 \\ -2i & 0 \end{pmatrix} \cdot \frac{1}{2}$$

$$= \begin{pmatrix} +2 & -2 \\ -2i & 0 \end{pmatrix} \cdot \frac{1}{2} = \begin{pmatrix} +1 & -1 \\ -i & 0 \end{pmatrix}$$

3. (15 pts) Show that there is no linear transformation sending the column vectors of $\begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 4 \\ 1 & 2 & 0 \end{pmatrix}$ to the column vectors of $\begin{pmatrix} 1 & 0 & 2 \\ 3 & 2 & 3 \\ 0 & 1 & 0 \end{pmatrix}$.

$$\text{rk} \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 4 \\ 1 & 2 & 0 \end{pmatrix} = \text{rk} \begin{pmatrix} 1 & 2 & 0 \\ 0 & -3 & 4 \\ 0 & 0 & 0 \end{pmatrix} = 2$$

$$\text{rk} \begin{pmatrix} 1 & 0 & 2 \\ 3 & 2 & 3 \\ 0 & 1 & 0 \end{pmatrix} = \text{rk} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & -3 \\ 0 & 1 & 0 \end{pmatrix} = \text{rk} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & -3 \\ 0 & 1 & 0 \end{pmatrix} = 3$$

The rank of the image of a LT is $\leq$ to the rank of the "source". So there is no such LT.

4. (15 pts) Define unitary matrices and show that if $U \in M_{n,n}(\mathbb{C})$ is a unitary matrix then $\|Uz\| = \|z\|$ for any $z \in \mathbb{C}^n$.

U is unitary if $U^H \cdot U = I$. In other words if u_j is the j -th column then $\bar{u}_j \cdot u_k = 0$ (or $\bar{u}_j^T u_k = 0$) or $u_j^H u_k = 0$ for all $j \neq k$ & $\bar{u}_j \cdot u_j = 1$.

$$\begin{aligned} \text{We have } \|z\|^2 &= z^H \cdot z = z^H I z = z^H U^H U z = (Uz)^H (Uz) \\ &= \|Uz\|^2 \end{aligned}$$

5. (20 pts) Define Hermitian matrices and show that the eigenvalues of a Hermitian matrix are real.

A is Hermitian if $A^H = A$ where $A^H = \overline{A}^T$ is the conjugate transposed matrix.

Suppose $Ax = \lambda x$, then $x^H Ax = x^H \lambda x = \lambda x^H x = \lambda \|x\|^2$
 $(A^H x)^H x = (Ax)^H x = (\lambda x)^H x = \overline{\lambda} x^H x = \overline{\lambda} \|x\|^2$

Since $\|x\|^2 > 0$ then $\lambda = \overline{\lambda}$ so $\lambda \in \mathbb{R}$.

6. (15 pts) What is the Jordan canonical form of $\begin{pmatrix} i & 0 & 1 & 0 \\ 0 & i & 0 & 2 \\ 0 & 0 & i & 1 \\ 0 & 0 & 0 & i \end{pmatrix}$?

$P = (i - \lambda)^4$ so $\lambda = i$ (4 times)

$N \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \middle| \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle$ is 2 dim $\Rightarrow$ We have 2 ones.

$B = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ $B^2 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ $B^3 = 0 \Rightarrow \begin{pmatrix} i & 1 & 0 & 0 \\ 0 & i & 1 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & i \end{pmatrix}$ JCF
 $\Downarrow$
 There is a Jordan block of length 3
 No Jordan block of length 4